We now show that the collection of all permutations of a nonempty set A forms a group under this permutation multiplication.

4.8 Theorem Let A be a nonempty set, and let S_A be the collection of all permutations of A. Then S_A is a group under permutation multiplication.

Proof We have shown that composition of two permutations of A yields a permutation of A, so S_A is closed under permutation multiplication.

Now permutation multiplication is defined as function composition, and in Section 1, we showed that *function composition is associative*. Hence $\mathscr{G}_1$ is satisfied.

The permutation ι such that $\iota(a) = a$, for all $a \in A$ acts as identity. Therefore $\mathscr{G}_2$ is satisfied.

For a permutation σ, the inverse function, σ^{-1}, is the permutation that reverses the direction of the mapping σ, that is, $\sigma^{-1}(a)$ is the element a' of A such that $a = \sigma(a')$. The existence of exactly one such element a' is a consequence of the fact that, as a function, σ is both one-to-one and onto. For each $a \in A$ we have

$$\iota(a) = a = \sigma(a') = \sigma(\sigma^{-1}(a)) = (\sigma\sigma^{-1})(a)$$

and also

$$\iota(a') = a' = \sigma^{-1}(a) = \sigma^{-1}(\sigma(a')) = (\sigma^{-1}\sigma)(a'),$$

so that $\sigma^{-1}\sigma$ and $\sigma\sigma^{-1}$ are both the permutation ι. Thus $\mathscr{G}_3$ is satisfied. ◆

Warning: Some texts compute a product $\sigma\mu$ of permutations in left-to-right order, so that $(\sigma\mu)(a) = \mu(\sigma(a))$. Thus the permutation they get for $\sigma\mu$ is the one we would get by computing $\mu\sigma$. Exercise 34 asks us to check in two ways that we still get a group. If you refer to another text on this material, be sure to check its order for permutation multiplication.

There was nothing in our definition of a permutation to require that the set A be finite. However, most of our examples of permutation groups will be concerned with permutations of finite sets. Note that the *structure* of the group S_A is concerned only with the number of elements in the set A, and not what the elements in A are. If sets A and B have the same cardinality, then $S_A \simeq S_B$. To define an isomorphism $\phi : S_A \to S_B$, we let $f : A \to B$ be a one-to-one function mapping A onto B, which establishes that A and B have the same cardinality. For $\sigma \in S_A$, we let $\phi(\sigma)$ be the permutation $\bar{\sigma} \in S_B$ such that $\bar{\sigma}(f(a)) = f(\sigma(a))$ for all $a \in A$. To illustrate this for $A = \{1, 2, 3\}$ and $B = \{\#, \$, \%\}$ and the function $f : A \to B$ defined as

$$f(1) = \#, \quad f(2) = \$, \quad f(3) = \%,$$

ϕ maps

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \text{ into } \begin{pmatrix} \# & \$ & \% \\ \% & \$ & \# \end{pmatrix}.$$

We simply rename the elements of A in our two-row notation by elements in B using the renaming function f, thus renaming elements of S_A to be those of S_B. We can take $\{1, 2, 3, \cdots, \mathrm{n}\}$ to be a prototype for a finite set A of n elements.

4.9 Definition Let A be the finite set $\{1, 2, \cdots, n\}$. The group of all permutations of A is the **symmetric group on n letters**, and is denoted by S_n. ■

Note that S_n has $n!$ elements, where

$$n! = n(n-1)(n-2)\cdots(3)(2)(1).$$

Let $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$. Then

$$\sigma\tau(1) = \sigma(1) = 2$$

and

$$\tau\sigma(1) = 3$$

which says that $\sigma\tau \neq \tau\sigma$. Therefore S_3 is not abelian. We have seen that any group with at most four elements is abelian. Furthermore we will see later that up to isomorphism, the abelian group $\mathbb{Z}_5$ is the only group of order 5. Thus S_3 is the smallest group which is not abelian.

4.10 Example Suppose that $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 1 & 4 & 5 \end{pmatrix}$. We find the inverse σ^{-1}. We saw in the proof of Theorem 4.8 that the inverse function of a permutation is the group inverse. So it is easy to find inverses for permutations, we simply turn the tables! That is, we switch the top and bottom rows and sort the columns so the top row is in order:

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 2 & 1 & 5 & 6 & 3 \end{pmatrix}. \quad \blacktriangle$$

Disjoint Cycles

There is a more efficient way of specifying the action of a permutation. In the two-row notation that we have been using, we list each number 1 through n twice, once in the top row and once in the bottom row. Disjoint cycle notation allows us to write the permutation using each number only once. We illustrate with an example. Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$. To write in disjoint cycle notation we start by writing

$$(1$$

We see that $\sigma(1) = 3$, so we place 3 just to the right of 1:

$$(1, 3$$

Now we see that σ maps 3 to 6, so we write:

$$(1, 3, 6$$

Our permutation maps 6 to 1, but there is no reason to write 1 again, so we just place a parenthesis after the 6 to indicate that 6 maps back to the first element listed:

$$(1, 3, 6)$$

This is called a **cycle** because when we apply σ repeatedly, we cycle through the numbers 1, 3, and 6. A cycle containing exactly k numbers is called a ***k*-cycle**. So the cycle $(1, 3, 6)$ is a 3-cycle. This is not the end of the story for σ because we have not indicated that 2 maps to 4. So we start another cycle and write

$$(1, 3, 6)(2, 4$$

to indicate that σ maps 2 to 4. Since 4 maps back to 2, we obtain a 2-cycle:

$$(1, 3, 6)(2, 4)$$

We still have not indicated what σ does to 5. We can write $(1, 3, 6)(2, 4)(5)$ to indicate that 5 maps to itself, but usually we will simply leave out 1-cycles with the understanding that any number not listed maps to itself. So in disjoint cycle notation

$$\sigma = (1, 3, 6)(2, 4).$$

We see that σ is a product of a 3-cycle and a 2-cycle. Sometimes we refer to a 2-cycle as a **transposition**.

A collection of cycles is said to be **disjoint** if no entry is in more than one cycle. Note that σ could also be written as $(3,6,1)(4,2)$, $(2,4)(1,3,6)$, or in a number of other ways. In general it doesn't matter which order we write the disjoint cycles, and inside each cycle we can start with any number as long as we keep the cyclic order the same. It is clear that any permutation in S_n can be written in disjoint cycle notation and that the representation is unique up to the order the cycles are written and the cyclic order within each cycle.

4.11 Example In disjoint cycle notation, $\sigma \in S_9$ is written as $(1,5,2,7)(3,4,9)$. Let us rewrite σ in two-row notation. Reading off the disjoint cycle notation we see that $\sigma(1) = 5$, $\sigma(5) = 2$, $\sigma(2) = 7$, $\sigma(7) = 1$, $\sigma(3) = 4$, $\sigma(4) = 9$, and $\sigma(9) = 3$. Since 6 and 8 do not appear in either cycle, we know that $\sigma(6) = 6$ and $\sigma(8) = 8$. Therefore,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 7 & 4 & 9 & 2 & 6 & 1 & 8 & 3 \end{pmatrix}$$ ▲

The operation that makes S_n a group is composition of functions. Keeping this in mind, we can see how to multiply permutations written in disjoint cycle notation.

4.12 Example Let $\sigma = (1,5,3,2,6)$ and $\tau = (1,2,4,3,6)$ in S_6. Let us find $\sigma\tau$ in disjoint cycle notation without resorting to using two-row notation. So

$$\sigma\tau = (1,5,3,2,6)(1,2,4,3,6).$$

We need to rewrite this product in disjoint cycles. So we ask where 1 is mapped. Since the operation is function composition, we see that the cycle τ on the right sends 1 to 2 and then the cycle on the left sends 2 to 6. So $\sigma\tau(1) = 6$ and we start our cycle by writing

$$(1,6$$

Now we see that τ maps 6 to 1 and σ maps 1 to 5, so we write

$$(1,6,5$$

We note that 5 is not in the cycle $(1,2,4,3,6)$, so $\tau(5) = 5$ and $\sigma\tau(5) = \sigma(5) = 3$. So we write

$$(1,6,5,3$$

Continuing in the same manner, we see that 3 maps to 1 and we complete the first cycle:

$$(1,6,5,3)$$

We are now ready to start the second cycle. We note that we have still not seen where 2 maps, so we start the next cycle with 2 and we write

$$\sigma\tau = (1,5,3,2,6)(1,2,4,3,6) = (1,6,5,3)(2,4)$$

using the same method we used for the first cycle. We know we are through since we have used every number 1 through 6. ▲

Example 4.12 illustrates the process of multiplying permutations in general. We move from right to left between the cycles, and within the cycles we move from left to right.

4.13 Example We compute the product of the permutations

$$\sigma = (1,5)(2,4)(1,4,3)(2,5)(4,2,1)$$

using disjoint cycle notation.

We start by seeing where 1 is mapped. The first cycle on the right maps 1 to 4. We are using function composition, so we next check what $(2,5)$ does to 4, which is nothing. So we move to the cycle $(1,4,3)$ and note that 4 is mapped to 3. Next, 3 is not in the cycle $(2,4)$ and so $(2,4)$ does not move 3 . Finally, $(1,5)$ also does not move 3 and we conclude that $\sigma(1) = 3$. We next need to determine where 3 is mapped by σ and continue until we arrive at

$$\sigma = (1,3,5,4)(2) = (1,3,5,4). \qquad \blacktriangle$$

It is interesting to note that in Example 4.13 the group was never specified. The same calculation is valid whether the group is S_5, S_6, or S_n for any $n \geq 5$.

4.14 Example We compute the inverse of $\sigma = (1,5,7)(3,8,2,4,6)$. We first note that in general for a group $(ab)^{-1} = b^{-1}a^{-1}$, so

$$\sigma^{-1} = (3,8,2,4,6)^{-1}(1,5,7)^{-1}.$$

The inverse of a cycle is simply the cycle written backward:

$$\sigma^{-1} = (6,4,2,8,3)(7,5,1).$$

This is a perfectly good way of writing σ^{-1}, but since disjoint cycles commute and we can start each cycle with any entry in the cycle, we could write

$$\sigma^{-1} = (1,7,5)(2,8,3,6,4). \qquad \blacktriangle$$

With a little practice, computing products of permutations in disjoint cycle notation becomes routine. We give the table for S_3.

4.15 Table

S_3 $\circ$	ι	$(1,2,3)$	$(1,3,2)$	$(1,2)$	$(1,3)$	$(2,3)$
ι	ι	$(1,2,3)$	$(1,3,2)$	$(1,2)$	$(1,3)$	$(2,3)$
$(1,2,3)$	$(1,2,3)$	$(1,3,2)$	ι	$(1,3)$	$(2,3)$	$(1,2)$
$(1,3,2)$	$(1,3,2)$	ι	$(1,2,3)$	$(2,3)$	$(1,2)$	$(1,3)$
$(1,2)$	$(1,2)$	$(2,3)$	$(1,3)$	ι	$(1,3,2)$	$(1,2,3)$
$(1,3)$	$(1,3)$	$(1,2)$	$(2,3)$	$(1,2,3)$	ι	$(1,3,2)$
$(2,3)$	$(2,3)$	$(1,3)$	$(1,2)$	$(1,3,2)$	$(1,2,3)$	ι

Again we can see that S_3 is not abelian since the table is not symmetric about the main diagonal. We also notice that although disjoint cycles commute, the same cannot be said for cycles that are not disjoint. For example we see in Table 4.15 that $(1,2)(2,3) = (1,2,3) \neq (1,3,2) = (2,3)(1,2)$.

The Dihedral Group

We next define a collection of finite groups based on the symmetries of regular n-gons. To be specific, we use as our standard regular n-gon the one whose points are U_n. Recall that U_n includes the point $(1,0)$ and the other points are spaced uniformly around the unit circle to form the vertices of a regular n-gon, which we denote by P_n. We label the points starting at $(1,0)$ with 0 and continue labeling them $1, 2, 3, \ldots, n-1$ around the circle counterclockwise. Note that this is the same labeling as the isomorphism between U_n and $\mathbb{Z}_n$ that we saw in Section 3. When we refer to a vertex we will reference it by its label. So vertex 0 is the point $(1,0)$. Note that the edges of P_n consist of the line segments between vertices k and $k +_n 1$ for $0 \leq k \leq n-1$.

8.17 Example In S_n for $n \geq 2$, the identity permutation is the product (1, 2) (1, 2) of transpositions. ▲

We have seen that every permutation of a finite set with at least two elements is a product of transpositions. The transpositions may not be disjoint, and a representation of the permutation in this way is not unique. For example, we can always insert at the beginning the transposition (1, 2) twice, because (1, 2) (1, 2) is the identity permutation. What is true is that the number of transpositions used to represent a given permutation must either always be even or always be odd. This is an important fact. The proof involves counting orbits and was suggested by David M. Bloom.

Let $\sigma \in S_A$ and $a \in A$. We let the **orbit** of a be the set $\{\sigma^k(a) \mid k \in \mathbb{Z}\}$. In the case of $\sigma \in S_n$, a simple way to think of the orbit of a is to think of the elements in the cycle containing a in the disjoint cycle representation of σ.

8.18 Example Let $\sigma = (1, 2, 6)(3, 5) \in S_6$. Then the orbit of 1 is the set $\{1, 2, 6\}$, which is also the orbit of 2 and the orbit of 6. The set $\{3, 5\}$ is the orbit of 3 and the orbit of 5. What about the orbit of 4? Recall that if we include 1-cycles, $\sigma = (1, 2, 6)(3, 5)(4)$, which says the orbit of 4 is $\{4\}$. ▲

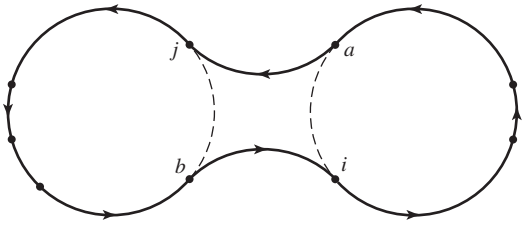
8.19 Theorem No permutation in S_n can be expressed both as a product of an even number of transpositions and as a product of an odd number of transpositions.

Proof Let $\sigma \in S_n$ and let $\tau = (i, j)$ be a transposition in S_n. We claim that the number of orbits of σ and of $\tau\sigma$ differ by 1.

Case I Suppose i and j are in different orbits of σ. Write σ as a product of disjoint cycles, the first of which contains j and the second of which contains i, symbolized by the two circles in Fig. 8.20. We may write the product of these two cycles symbolically as

$$(b, j, \times, \times, \times)(a, i, \times, \times)$$

where the symbols $\times$ denote possible other elements in these orbits.



8.20 Figure

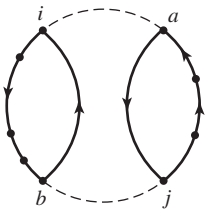
Computing the product of the first three cycles in $\tau\sigma = (i, j)\sigma$, we obtain

$$(i, j)(b, j, \times, \times, \times)(a, i, \times, \times) = (a, j, \times, \times, \times, b, i, \times, \times).$$

The original 2 orbits have been joined to form just one in $\tau\sigma$ as symbolized in Fig. 8.20. Exercise 42 asks us to repeat the computation to show that the same thing happens if either one or both of i and j should be the only element of their orbit in σ.

Case II Suppose i and j are in the same orbit of σ. We can then write σ as a product of disjoint cycles with the first cycle of the form

$$(a, i, \times, \times, \times, b, j, \times, \times)$$



8.21 Figure

shown symbolically by the circle in Fig. 8.20. Computing the product of the first two cycles in $\tau\sigma = (i,j)\sigma$, we obtain

$$(i,j)(a,i,\times,\times,\times,b,j,\times,\times) = (a,j,\times,\times)(b,i,\times,\times,\times).$$

The original single orbit has been split into two as symbolized in Fig. 8.21.

We have shown that the number of orbits of $\tau\sigma$ differs from the number of orbits of σ by 1. The identity permutation ι has n orbits, because each element is the only member of its orbit. Now the number of orbits of a given permutation $\sigma \in S_n$ differs from n by either an even or an odd number, but not both. Thus it is impossible to write

$$\sigma = \tau_1\tau_2\tau_3\cdots\tau_m\iota$$

where the τ_k are transpositions in two ways, once with m even and once with m odd. ◆

8.22 Definition A permutation of a finite set is **even** or **odd** according to whether it can be expressed as a product of an even number of transpositions or the product of an odd number of transpositions, respectively. ■

8.23 Example The identity permutation ι in S_n is an even permutation since we have $\iota = (1,2)(1,2)$. If $n = 1$ so that we cannot form this product, we define ι to be even. On the other hand, the permutation (1, 4, 5, 6) (2, 1, 5) in S_6 can be written as

$$(1,4,5,6)(2,1,5) = (1,6)(1,5)(1,4)(2,5)(2,1)$$

which has five transpositions, so this is an odd permutation. ▲

The Alternating Groups

We claim that for $n \geq 2$, the number of even permutations in S_n is the same as the number of odd permutations; that is, S_n is split equally and both numbers are $(n!)/2$. To show this, let A_n be the set of even permutations in S_n and let B_n be the set of odd permutations for $n \geq 2$. We proceed to define a one-to-one function from A_n onto B_n. This is exactly what is needed to show that A_n and B_n have the same number of elements.

Let τ be any fixed transposition in S_n; it exists since $n \geq 2$. We may as well suppose that $\tau = (1,2)$. We define a function

$$\lambda_\tau : A_n \to B_n$$

by

$$\lambda_\tau(\sigma) = \tau\sigma,$$

that is, $\sigma \in A_n$ is mapped into $(1,2)\sigma$ by λ_τ. Observe that since σ is even, the permutation $(1,2)\sigma$ can be expressed as a product of a (1 + even number), or odd number, of transpositions, so $(1,2)\sigma$ is indeed in B_n. If for σ and μ in A_n it is true that $\lambda_\tau(\sigma) = \lambda_\tau(\mu)$, then

$$(1,2)\sigma = (1,2)\mu,$$

and since S_n is a group, we have $\sigma = \mu$. Thus λ_τ is a one-to-one function. Finally,

$$\tau = (1,2) = \tau^{-1},$$

so if $\rho \in B_n$, then

$$\tau^{-1}\rho \in A_n,$$

and

$$\lambda_\tau(\tau^{-1}\rho) = \tau(\tau^{-1}\rho) = \rho.$$

Thus λ_τ maps onto B_n. Hence the number of elements in A_n is the same as the number in B_n since there is a one-to-one correspondence between the elements of the sets.

Note that the product of two even permutations is again even. Also since $n \geq 2$, S_n has the transposition $(1, 2)$ and $\iota = (1, 2)(1, 2)$ is an even permutation. Finally, note that if σ is expressed as a product of transpositions, the product of the same transpositions taken in just the opposite order is σ^{-1}. Thus if σ is an even permutation, σ^{-1} must also be even. Referring to Theorem 5.12, we see that we have proved the following statement.

8.24 Theorem If $n \geq 2$, then the collection of all even permutations of $\{1, 2, 3, \cdots, n\}$ forms a subgroup of order $n!/2$ of the symmetric group S_n.

We can define a function called the **sign of a permutation**, $\text{sgn} : S_n \to \{1, -1\}$ by the formula

$$\text{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Thinking of $\{1, -1\}$ as a group under multiplication, it is easy to see that sgn is a homomorphism. Since 1 is the identity in the group $\{1, -1\}$, $\text{Ker}(\text{sgn}) = \text{sgn}^{-1}[\{1\}]$ is a subgroup of S_n consisting of all the even permutations. The homomorphism sgn is used in the standard way of defining the determinant of a square matrix. Exercise 52 asks you to prove some of the standard facts about determinants using this definition.

8.25 Definition The subgroup of S_n consisting of the even permutations of n letters is the **alternating group A_n on n letters**. ■

Both S_n and A_n are very important groups. Cayley's theorem shows that every finite group G is structurally identical to some subgroup of S_n for $n = |G|$. It can be shown that there are no formulas involving just radicals for solution of polynomial equations of degree n for $n \geq 5$. This fact is actually due to the structure of A_n, surprising as that may seem!

EXERCISES 8

Computations

In Exercises 1 through 10 determine whether the given map is a group homomorphism. [Hint: To verify that a map is a homomorphism, you must check the homomorphism property. To check that a map is not a homomorphism you could either find a and b such that $\phi(ab) \neq \phi(a)\phi(b)$, or else you could determine that any of the properties in Theorem 8.5 fail.]

1. Let $\phi : \mathbb{Z}_{10} \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder when x is divided by 2.
2. Let $\phi : \mathbb{Z}_9 \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder when x is divided by 2.
3. Let $\phi : \mathbb{Q}^* \to \mathbb{Q}^*$ be given by $\phi(x) = |x|$.
4. Let $\phi : \mathbb{R} \to \mathbb{R}^+$ be given by $\phi(x) = 2^x$.
5. Let $\phi : D_4 \to \mathbb{Z}_4$ be given by $\phi(\rho^i) = \phi(\mu\rho^i) = i$ for $0 \leq i \leq 3$.